

CHECK WHAT YOU LEARN'T

Check Yourself - Unit 2

A. Choose the correct answer

- The actual computer on an Arduino Uno is
(a) the USB chip (b) the ATmega328P (c) the regulator (d) the crystal
- Variables live in
(a) flash (b) SRAM (c) EEPROM (d) the ADC
- A 10-bit ADC reports
(a) 10 values (b) 100 values (c) 1024 values (d) 1023 volts
- One ADC step across a 5V reference is about
(a) 4.9 microvolts (b) 4.9 millivolts (c) 49 millivolts (d) 0.5 volts
- The ADC finds its answer by
(a) counting up from zero (b) successive approximation (c) measuring current (d) guessing
- A millis() variable must be declared as
(a) int (b) float (c) unsigned long (d) boolean

B. Fill in the blanks

- The _____ receives your code and writes it into flash.
- The ATmega328P has _____ KB of flash and _____ KB of SRAM.
- delay() is unacceptable in a real system because it _____ the processor.
- Each ADC conversion takes roughly _____ microseconds.

C. Answer in one or two lines

- Describe the four stages between pressing Upload and your program running.
- A pin sits at 1.8 volts. What will analogRead() return? Show your working.
- Explain why now - lastRead is safer than now > lastRead + 100.
- Give one concrete example of a fault caused by using delay() in a moving robot.

UNIT 3

Basic Project: Laser Security System

Sessions 6, 7 and 8

A beam of light across a doorway, a receiver that notices when it breaks, and a counter that remembers every intrusion. Simple to describe, and full of the engineering decisions that separate a demonstration from a product.

By the end of this unit you will be able to:

- Explain how a laser diode produces a collimated beam.
- Determine a detection threshold from measured data rather than trial and error.
- Implement latching so that a fast event is never missed.
- Produce a written test report with false-positive and false-negative counts.

SESSION 6

one class

Laser Diodes, LDRs and Optical Alignment

Today you will

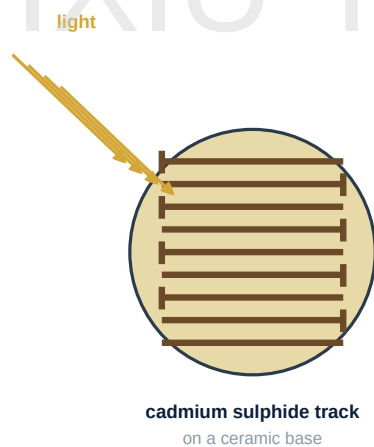
- Learn how a laser differs from an ordinary LED.
- Understand what is inside an LDR and why it responds to light.
- Align the beam and record threshold data.

A laser diode starts life as an ordinary light-emitting junction, but two additions transform it. First, the semiconductor is built as an **optical cavity** - mirrored at both ends, so light bounces back and forth and stimulates more light of exactly the same wavelength and exactly in step. Second, a tiny lens at the front **collimates** the output into a narrow parallel beam instead of letting it spread.

The result is what makes the whole project possible: an ordinary LED at three metres would spread into a dim patch, while a laser still arrives as a bright dot small enough to fall entirely on a sensor the size of your fingernail.

	Ordinary LED	Laser diode
Wavelengths emitted	a spread of colours	essentially one
Phase of the waves	random	in step with each other
Beam spread	wide, tens of degrees	narrow, fractions of a degree
Useful range for this project	a few centimetres	several metres
Eye safety	harmless	never look into the beam

The receiver is your LDR. Under its clear window is a zig-zag track of **cadmium sulphide** printed on ceramic. The zig-zag makes the conducting path long and narrow, which magnifies the effect you are about to exploit.



What the light actually does

In the dark

electrons stay locked in place
resistance high, about 1,000,000 ohms

In bright light

photons knock electrons free
resistance low, a few hundred ohms

This is called photoconductivity. Notice what the sensor does not do: it never measures brightness in any unit. It only changes its resistance, and the module turns that into a voltage your Arduino can read.

Figure - Inside an LDR: light knocks electrons loose and current flows more easily

Behind the Scenes

In darkness the electrons in cadmium sulphide are bound in place and the resistance can reach a million ohms. Each arriving photon carries just enough energy to free one electron, and every freed electron makes the material a little more conductive. Under a laser spot the resistance collapses to a few hundred ohms. The effect is called **photoconductivity**. Note the consequence: an LDR is slow, taking tens of milliseconds to settle - fine for a doorway, useless for high-speed counting.

Try This

Fix the laser and the LDR facing each other about 40 cm apart, both clamped or taped down. Print `analogRead(A0)` continuously. Record the value with the beam landing squarely on the sensor, then with your hand interrupting it, then with the room lights off, and finally with a torch shone at the LDR. Those four numbers decide your threshold - and the last one will tell you how easily your system can be fooled.

Condition	Reading on A0	What it tells you
Beam landing on the sensor	_____	the armed value
Beam interrupted by a hand	_____	the alarm value
Room lights off, beam on	_____	how much ambient light contributes
Torch shone at the LDR, beam broken	_____	how a defeat attempt would look
Threshold you chose	_____	justify it from the four above

SESSION 7

one class

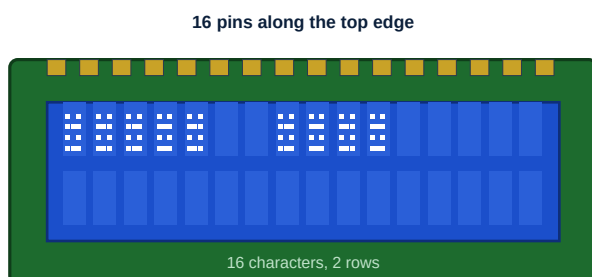
Latching Alarms and the LCD Counter

Today you will

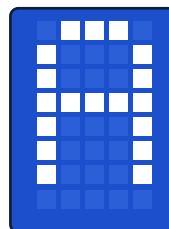
- Wire the laser, LDR, display, buzzer and reset switch.
- Understand why a fast event needs latching.
- Show a running intrusion count on the display.

Before wiring, it is worth knowing what the display in front of you actually is. The name sounds like a contradiction: how can something be both liquid and crystal? A liquid crystal is a genuine fourth state of matter - its long molecules flow freely like a liquid, yet they also line up with one another like a crystal, and that alignment can be changed by a small voltage.

Inside a 16x2 character LCD



a 5 x 8 grid of dots per character



the HD44780 controller stores the dot pattern for 208 characters

Why it is called liquid crystal

Between two sheets of glass sits a liquid whose long molecules normally line up in a twist. That twist rotates light passing through, so it escapes the far filter and the pixel looks pale. Apply a small voltage and the molecules straighten. The light no longer twists, the far filter blocks it, and that dot goes dark. No dot makes its own light - the backlight behind does that, which is why an LCD needs its LED backlight and why the contrast pot changes what you can read.

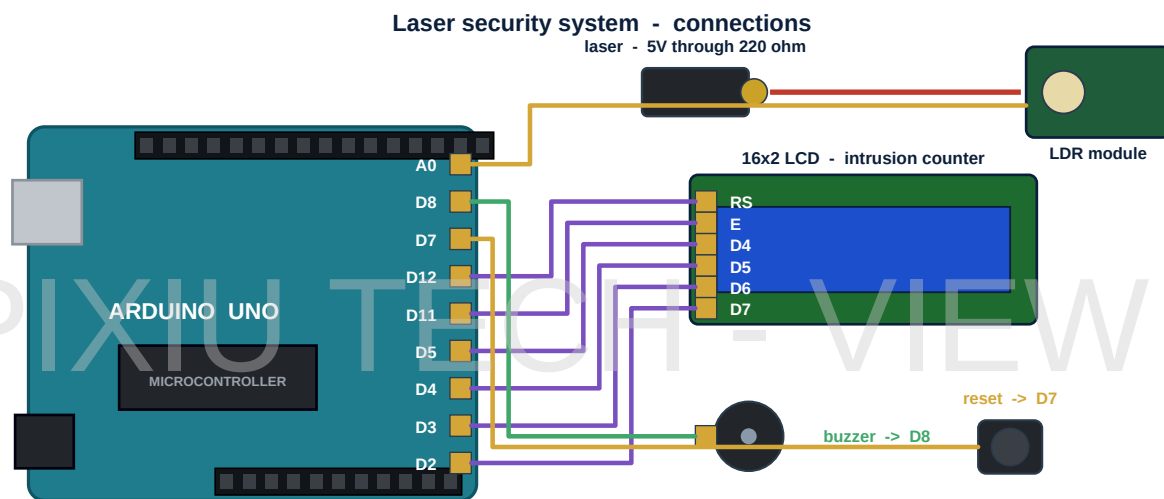
Behind the Scenes

An LCD dot produces **no light of its own** - it only blocks or passes light from the backlight behind it. That is why the display is unreadable if the backlight fails, why the contrast potentiometer matters so much, and why an LCD consumes so little power compared with a screen that generates its own light. The black blob on the back is an **HD44780** controller, the industry standard since the 1980s: it stores the dot patterns for 208 characters and refreshes every dot continuously, so you only ever send it a letter.

LCD pin	Name	Purpose
1, 2	VSS, VDD	Ground and 5V
3	V0	Contrast - this is what the potentiometer sets
4	RS	Register Select: is this byte a character or a command?
5	RW	Read or Write. Tied to ground - we only ever write.
6	E	Enable. A pulse means 'take the data now'.
7 to 14	D0 to D7	Data. In 4-bit mode only D4 to D7 are used.
15, 16	A, K	LED backlight. A to 5V through 220 ohm.

Behind the Scenes

Why use four data lines instead of eight? Because eight would consume eight Arduino pins and leave nothing for your sensors. In **4-bit mode** each byte is sent as two halves. It is half the speed - and entirely irrelevant, since the display needs updating a few times a second while the chip runs sixteen million times faster.



The laser is powered continuously. The Arduino only watches the LDR.

Figure 3.2 - A beam across a doorway, and a counter that remembers

Component	Arduino pin	Note
Laser module +	5V through 220 ohm	Powered continuously, not switched
Laser module -	GND	
LDR module AO	A0	The beam intensity
Buzzer +	D8	The alarm
Push switch	D7, with 10K to GND	Clears the latch
LCD RS / E	D12 / D11	Control lines
LCD D4-D7	D5, D4, D3, D2	Data, in that order
LCD V0	Potentiometer wiper	Contrast
LCD VSS, RW, K	GND	
LCD VDD, A	5V (A through 220 ohm)	Power and backlight

Now the design decision that matters. Somebody walking briskly through the beam interrupts it for perhaps eighty milliseconds. If your loop happens to be busy writing to the display at that instant, a simple if statement would miss the event entirely - and an intruder detector that misses intruders is worthless.

The answer is **latching**. A boolean records that the beam was broken. Once set, only the reset button can clear it. The machine has stopped merely reacting and started remembering.

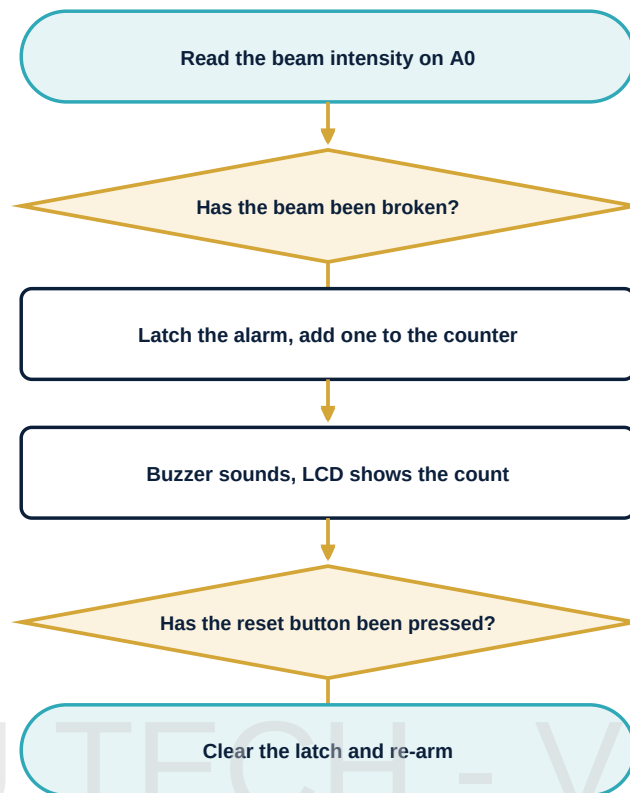


Figure 3.3 - Detect, latch, count, and wait for a human

SESSION 8

one class

Reset, False Triggers and a Written Test Report

Today you will

- Complete the program with debounced reset and timestamps.
- Run a formal test and count false positives and negatives.
- Write a one-page test report.

LaserSecuritySystem.ino

```

#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

int ldrPin = A0, buzzer = 8, resetBtn = 7;
int threshold = 500;           // set this from YOUR measured data

boolean alarmLatched = false;
int intrusions = 0;
unsigned long lastEvent = 0;
unsigned long lastBeep = 0;
boolean beepState = false;

void setup() {
  pinMode(buzzer, OUTPUT);

```

```

pinMode(resetBtn, INPUT);
lcd.begin(16, 2);
lcd.print("System armed");
Serial.begin(9600);
}

void showStatus() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(alarmLatched ? "INTRUSION!" : "System armed");
  lcd.setCursor(0, 1);
  lcd.print("Count: ");
  lcd.print(intrusions);
  if (intrusions > 0) {
    lcd.print(" ");
    lcd.print((millis() - lastEvent) / 1000); // seconds since last
    lcd.print("s");
  }
}

void loop() {
  unsigned long now = millis();
  int beam = analogRead(ldrPin);

  if (beam < threshold && !alarmLatched) { // beam broken, first time
    alarmLatched = true;
    intrusions = intrusions + 1;
    lastEvent = now;
    Serial.print("Intrusion ");
    Serial.print(intrusions);
    Serial.print(" at ");
    Serial.println(now);
    showStatus();
  }

  if (digitalRead(resetBtn) == HIGH) { // only a human clears it
    alarmLatched = false;
    digitalWrite(buzzer, LOW);
    showStatus();
    delay(300); // simple debounce
  }

  if (alarmLatched && now - lastBeep > 200) {
    lastBeep = now;
    beepState = !beepState;
    digitalWrite(buzzer, beepState); // non-blocking beeping
  }

  if (!alarmLatched && now - lastEvent > 1000 && now % 1000 < 20) {
    showStatus(); // refresh the idle display
  }
}

```

Three professional details are hiding in that code. The buzzer beeps using `millis()` rather than `delay`, so the system can still detect a second intrusion while it is already alarming. The condition includes `&& !alarmLatched` so one continuous interruption is counted once, not five hundred times. And every event is timestamped to the serial line, which is what turns a demonstration into evidence.

8.1 Testing Formally

A detector can fail in two distinct ways and they are counted separately, because their consequences differ completely. A **false positive** is an alarm with no intruder - annoying, and eventually fatal to trust. A **false negative** is an intruder with no alarm - the system has failed at the only job it had.

Test	Runs	Alarms raised	Result
Undisturbed for 5 minutes	1	_____	0 expected
Hand passed through the beam	10	_____	10 expected
Object passed quickly, under 100 ms	10	_____	10 expected
Room lights switched off and on	5	_____	0 expected
Torch shone at the LDR, beam broken	5	_____	note what happens
False positives		_____	
False negatives		_____	

Behind the Scenes

The torch test is the interesting one. If a bright torch on the LDR keeps the reading above your threshold even while the beam is blocked, then your alarm can be defeated by anybody with a phone. Real beam-break sensors defeat this by **modulating** the beam - pulsing it at a known frequency and accepting only light that pulses in the same rhythm. Steady light, however bright, is ignored.

Challenge

Write the one-page test report: a paragraph of purpose, your threshold and the data that justified it, the table above completed, the false positive and negative counts, and two sentences on the limitations you found. This document is assessed alongside the hardware.

New word	What it means
Collimated	Formed into a narrow parallel beam.
Optical cavity	Mirrored ends that make a laser's light coherent.
Photoconductivity	Light freeing electrons and lowering resistance.
Latching	Staying on once triggered, until deliberately cleared.
Debounce	Ignoring the brief contact chatter of a switch.
Modulation	Pulsing a signal at a known rate so it cannot be imitated.